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LECTURE NOTE
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MANAGEMENT INFORMATION SYSTEM

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GENERAL SYSTEM THEORY

System

The word "System" is derived from Greek word *Systema*, which means an organized relationship between any set of components to achieve some common cause or objective. A system can be described as a group of interrelated elements that are integrated with the common purpose of achieving a particular objective by accepting inputs and producing outputs in an organized transformation process. By this definition, an organization such as a firm or a school is a system. The organization consists of the physical resources like; Manpower, Material, Machines, Money, and information (including data) and they work towards achieving *particular objectives* that are specified by the owners or management. Other examples of system are information systems, financial systems, ecological systems, computer systems, electrical system, education system, operating system, sound system, waste disposal system, etc.

A system can be "an orderly grouping of interdependent components linked together according to a plan to achieve a specific goal."

Sub System

A Subsystem is simply a system within a system. This means that systems exist on more than one level i.e. Systems can be composed of subsystems or elemental parts. For example, the system of the firm includes subsystems of the bank may be such departments as savings, demand deposit (checking accounts), and loan.

Super System

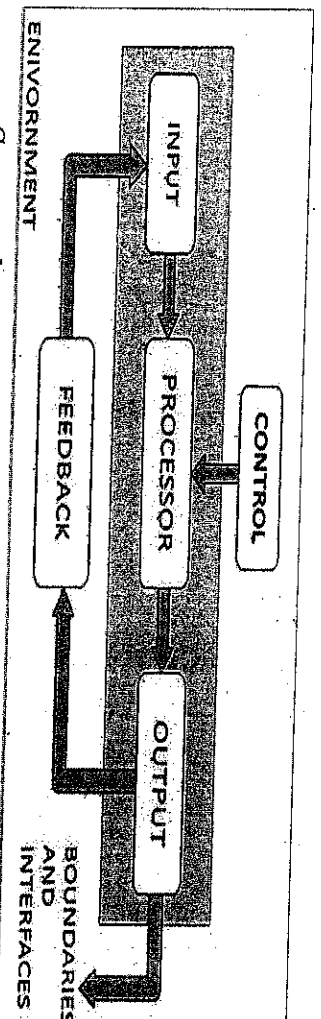
When a system is part of a larger system, the larger system is the super system. For example, the system of the firm exists in one or larger environmental systems or super systems. If the firm is a bank, for example, it is part of the financial community. It is also part of the business community, it is part of a local community, and the global community.

The manager's major responsibility is to ensure that the firm meets its objectives, by ensuring that the various parts of the firm/company work together as they should.

Basic Elements of System

A basic configuration of system consists of following elements;

- Input and output element.
- Transformation process.
- Feedback.
- Control mechanism.
- Objectives.
- Environment
- Boundaries



Conceptual view of the basic elements of a system

1. Input are the information that enters into the system, it involves capturing and assembling elements that are used for feeding the data to be processed. Examples are labour, raw materials, fuel or electricity, clerical data.
2. Processing involves transformation processes that convert input into meaningful output. Examples are mathematical calculations, and a manufacturing process. In the example of the car manufacturer, it combines labour, raw materials and other assets to turn raw materials into automobiles.
3. Output is basically the result of processing, it involves the elements that communicate the results of processing to the outside world or ultimate destination for the benefit of the users. Examples of these include management information, finished products, and human services.
4. Feedback is defined as data about the performance of a system. For example, data about exams performance is feedback to an exams officer. The information that you can derive from feedback enables you to alter your input requirements, such as new machinery, alternative materials, different production processes or more efficient technologies. Although it primarily affects your inputs, this feedback might also have an indirect effect on your transformation process.
5. Control involves monitoring and evaluating feedback to determine whether a system is moving towards the achievement of goals. The control function then makes necessary adjustments to a system's input and processing components to ensure that it produces the set objectives. The basic elements of Control mechanism are as follows:
 - i. Specification Planning: A standard specifying the expected performance must be set.
 - ii. Collecting facts: A measurement of the actual performance to check for accuracy, speed, and anticipated results.
 - iii. Comparison: This attempts to isolate the reason for any variation
 - iv. Corrective Action: This may be achieved through feedback.
6. Environment of a system is the setting or location in which it operates. This can dictate the resources it has access to, its relationships with other systems and even the nature of the organization's activities. If the organizational system itself is a business, then the environment might be the country, region, county or city in which it operates.
7. Boundary is a conceptual line that divides the system that you want to study from its environment. It is useful to think of a system's environment as being made up of those things that are not part of the system, but can either affect the system or be affected by it.

NB; Not all systems are able to control their own operations. A System without the control mechanism, feedback loop and objective elements is called *dynamic system*.

Classification of Systems

There are three categories of systems; Probabilistic system, cybernetic system and deterministic system.

1. Probabilistic system

This is a system whose outcome cannot be predicted with precision. That is, not certain what outputs will be achieved from specific input because these systems are subjected to random influences from the internal and external environment. Therefore, it cannot be predicted precisely. Examples include business system, economic system, agricultural system. It is also called stochastic system.

2. Deterministic System

This is system whose behavioral patterns can be predicted if its present state and operation characteristics are known. Such a system operates according to a predetermined set of rules. A good example of a deterministic system is a computer program. The computer programmer knows in advance the output his program will produce, if the program is error free. It also called mechanistic system.

3. Cybernetic System

A cybernetic System is one that automatically adapts to its environments. A system with feedback and control components is sometimes called a cybernetic system, that is, a self-monitoring, self-regulating system. The behavior of the cybernetic system is a subject of rigorous scientific investigation.

4. Adaptive Systems:

These are dynamic system which responds to changing circumstances by adjusting its behavior on a self-organizing basis. The system alters its parameters as a result of measuring output. Example includes organizational systems such as human beings. The degree to which a business is a adaptive depends to a great extent on the caliber of its management.

SYSTEM THEORY

The systems theory of management asserts that any organisation is a single, unified system of interrelated parts or subsystems. Each part of the overall system is dependent on the others and cannot function optimally without them.

Open System

This is a system that interacts with other systems by exchanging input and output components with its environmental elements such as customers, suppliers, competitors, the labour force, government and global community. A firm and most systems that occur naturally in business are examples of open system. It has to adapt to its changing environments where necessary in order to ensure its continued existence. Also, a business system must be capable of reorganization in order to cope with the changing market situations.

- Open system theory often refers to how an organizational structure affects the flow of information and interaction internally and with the external environment.
- Open system theory is a biological theory that has been adapted to business organizational structure theories. It may seem odd that biology has anything to do with business, but the theory recognizes that any system is assembled by many parts that share resources or data internally, and with its external environment.

Closed System

This is a system whose function and behavior is not affected by its environment. That is, it is self-contained and the input/output elements do not interface with environmental elements outside it. Example includes chemical reactions in a controlled laboratory etc, other examples of Closed systems are an automatic watch. Workers in closed systems within an organization don't communicate with other departments about their activities, nor do they receive input from other departments.

- Assembly line workers complete their daily tasks without having to worry about their work being interrupted by an event such as a staff meeting. Assembly line workers must also adhere to a rigid set of procedures that ensure efficiency and productivity.
- The research and development division of a business is a closed system, because the people responsible for developing new products or new technology work without consulting other divisions in the company.
- Some businesses maintain proprietary trade secrets that they must guard to prevent the competition from stealing this information.

General Nature of a System

There are essentially four (4) views that support the General System Concept. These include the following attributes:

- a.) **Synergy:** All systems are made up of subsystems and for a meaningful Management Information System, the subsystems must act as one body where the whole is the greater than sum of all the subsystems.

b.)**Hierarchy:** The system when viewed as a whole, is made up of subsystems in which control flows in an ordered sequence from the top to the bottom.

c.)**Interdependency:** All the parts of a system constitute an indissoluble whole so that no part can be altered without affecting the other parts.

d.)**Sub-optimality:** This is a situation in which the sub-systems become selfish and self-centered and pursuing their own objectives independently. The overall effect may be detrimental to the existence of the whole organization.

Other Characteristics of system include;

1. Systems have a specific structure which is defined by its components (entities/subsystems) and processes (interrelationships between its components). A system is a collection of interrelated entities and/or subsystems which can be analyzed. It is possible to understand the specific structure of a system. However, in some systems complete knowledge may not be available but in most cases the fundamental entities and their interrelations are known.
2. Systems are a model of reality, a system is an abstraction of reality. It is created to comprehend the nuances of a real-world condition and understand the interrelationships of subsystems in such real-world conditions in greater clarity.
3. A system has a purpose-a system performs a function. It has a reason for its existence. The purpose in most cases is the output of the system and in a way the output defines the purpose of the system.
4. Systems have inputs and outputs. Outputs are produced by processing the inputs-a system (unless of theoretical interest and fully closed), interacts with the environment by taking in input and then after processing the input produces the output.
5. Systems have performance that can be measured in terms of its output-a system will have measures of performance. In most cases, the performance of the system is a function of its input and output.
6. A system serves a client, the system will have a utility and hence, a client for it. The client can also be another system.
7. The components that make up a system have functional as well as structural interrelationships with each other.
8. A system has an environment-a system cannot exist in isolation. It exists in an environment. The environment reacts with it.

CONCEPT OF MANAGEMENT INFORMATION

Management is a universal phenomenon. It is a very popular and widely used term. All organizations - business, political, cultural or social are involved in management because it is the management which helps and directs the various efforts towards a definite purpose.

According to *Harold Koontz*, "Management is an art of getting things done through and with the people in formally organized groups. It is an art of creating an environment in which people can perform and individuals and can co-operate towards attainment of group goals".

According to *F.W. Taylor*, "Management is an art of knowing what to do, when to do and see that it is done in the best and cheapest way".

Management can be simply defined as the organization and mobilization of all human and material resources in a particular system for the achievement of identified objectives in the system. The two key words are 'human' and 'materials'. It is the business of management that both human and material resources within a system should be organized and mobilized so that the objectives of the system can be achieved to the fullest. Then administration or management therefore influences the results to be achieved, the direction to be pursued and the priorities to be reorganized.

Top Level Management

The board of directors, the CEO, and/or the managing director makes up top level of management. The top management oversees an organization's goals and policies and is the ultimate source of authority. It spends more time coordinating and planning tasks.

Middle Level Management

Middle level managers include branch managers and departmental managers. For the operation of their department, they answer to top management. They spend more time performing organizational and directional functions. There is just one layer of middle level management in small organizations, although middle level management may exist in large corporations.

Lower Level Management

A lower level of management is also referred to as the supervisory or operational level. Supervisors, foremen, section officers, superintendents, etc. make up lower level of management. R.C. Davis explains that "supervisory management refers to those executives whose work is mostly with personal control and direction of operational employees." In other words, they are focused on the managerial function of direction and control.

What is Information Systems?

An information system is a set of interrelated components that collects, stores, analyses and disseminates information for effective business decision-making. For example, an organisation wishes to increase its profits by boosting its sales.

CATEGORIES OF INFORMATION

There are three main categories of information and these are related to the purpose for which the information is utilized. The three categories of information are:

- i) **Strategic Information (Top management):** This is used by senior managers to plan the objectives of their organization, and to assess whether the objectives are being met in practice. Example is Board of Directors whose duties include to plan product range, plan expression into new markets and to establish at least 5-year plan and annual plans for the corporation. Information needed includes highly summarized information regarding corporation activities to-date e.g sales, transactions, Forecasts, awareness of competitor activities, and awareness of overall economic climate.
- ii) **Tactical Information (Middle management):** This is used by middle management to decide how the resources of the business should be employed and to monitor how they are being and have been employed. Example, Sales Managers whose duties include ensuring that sales target are met, determining monthly sales plan, and correcting variances from plan. Examples of information needed -monthly sales analysis, market conditions, competitors' activity.
- iii) **Operational Information (Low-level management):** This is used at the lower-level management for daily or hourly running of their section. Users of operational information include foremen, sectional heads, sales clerk etc, to ensure that specific tasks are planned and carried out properly within the factory or office. Information needed includes prices of items, customer's credit rating, stock availability etc.

Characteristics of Operational Information

- i) Derived almost entirely from internal sources.
- ii) It is highly detailed.
- iii) It relates to the immediate term.
- iv) It is task-specific.
- v) It is prepared constantly or very frequently.

Management Functions

These can be applied to organization, be it industry, education or government. These elements are:

- P=Planning
- O=Organizing
- D=Directing
- C=Co-coordinating
- C=Controlling

Planning: This is the first step in any administrative function. It involves making arrangements for the future. It deals on how to use time, personnel and equipment in future. The future may begin with days, weeks, month or year. Plans are made after objectives are set. Planning is usually regarded as a search for the best method of achieving these goals.

According to KOONTZ, "Planning is deciding in advance - what to do, when to do & how to do. It bridges the gap from where we are & where we want to be".

Planning is necessary to ensure proper utilization of human & non-human resources. It is all pervasive, it is an intellectual activity and it also helps in avoiding confusion, uncertainties, risks, wastages etc.

Organizing: According to Henry Fayol, "To organize a business is to provide it with everything useful or its functioning i.e. raw material, tools, capital and personnel's". The manager decides what jobs would have to be filled and the duties and responsibilities attached to each one. Effective organization implies that the right people are assigned to the correct jobs at the appropriate time. Activities should be assigned to individuals on the basis of their ascertained qualification, experience and interests. On the part of the manager, a lot of wisdom, maturity and objectivity are required.

Directing: is that inter-personnel aspect of management function which deals directly with influencing, guiding, supervising, and motivating sub-ordinate for the achievement of organizational goals. This involves instructing and giving staff details on how to go about their jobs. It is not once for all activity but an on-going one. workers have to be continually put on the right path.

Co-ordinating: This relates to the activities of different units to one another to ensure that the different activities are met, to achieve the set objectives. Coordination occurs when the organization's numerous activities are planned, organized, directed and controlled.

Controlling: It is the measuring and correcting of activities are contributing to the achievement of planned goals. Some authors have pointed out that appraisal, evaluating and supervision are the administrator's tools for controlling, directing and leading.

LEVELS OF MANAGEMENT

An organization's "Levels of Management" serve as a boundary between different managerial levels. When a business and its workforce grow in size, so do the levels of management, and vice versa. Any managerial position's authority, status, and chain of command are all based on the level of management.

There are three basic levels of management: Top Level Management, Middle Level Management, Lower Level Management

vi) It is largely quantitative.

Characteristics of Tactical Information

- i) It is relevant to short and medium term.
- ii) It describes or analyses activities or departments.
- iii) it is prepared routinely and regularly.
- iv) it is based on quantitative measures (cash flow forecasts, budgetary control or variance analysis reports)

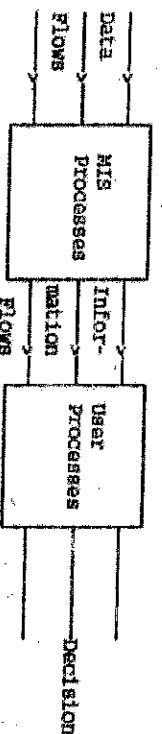
Characteristics of Strategic Information

- i) It is derived from both internal and external sources
- ii) It is usually summarized
- iii) It is relevant to the long term
- iv) It deals with the whole organisation
- v) it is both quantitative and qualitative (future market prospects, total cash needs, capital equipment)

MANAGEMENT INFORMATION SYSTEM (MIS)

A Management Information System (MIS) can be defined as an assemblage of data (e.g. facts and opinions) that are processed (e.g summarized, categorized, or projected) in such a way that they result in intelligent information that management can use to make decisions, and that the organization can use to attain its goals.

MIS can also be defined as the combination of human and computer based resources in the collection, storage, retrieval, communication and use of data for the purpose of efficient management and planning of a business enterprise



The Main Objectives of MIS is to use formalized procedures to provide management with appropriate information from all relevant sources, which would enable the manager make timely and effective decisions. The information must be provided at an acceptable level of accuracy and at an economical cost.

Organizational information systems could include the following:

- i. Marketing Information System,
- ii. Manufacturing Information System,
- iii. Financial Information System,
- iv. Human Resources Information System,
- v. Information Resources Information System.
- vi. Executive Information System.

STRUCTURE OF MANAGEMENT INFORMATION SYSTEM (MIS)

1. **Strategic Planning and control:** "strategic planning is the process of deciding on objectives of the organization, on changes in these objectives, on the resources used to attain these objectives, and on the policies that are to govern the acquisition, use, and disposition of these resources.
2. **Management control:** This is the process by which managers assure that resources are obtained and used effectively and efficiently in the accomplishment of the organization's objectives.
3. **Operational control and Planning:** This means "the process of assuring that specific tasks are carried out effectively and efficiently."

The basic difference between management control and operational control is that operational control is concerned with *tasks* (such as manufacturing a specific part) whereas management control is most often concerned with *people*.

The importance of MIS to business organizations

The importance derived from implementing a management information system includes the following:

1. Provision of improved services to customers.
2. It facilitates decision making process of managers.
3. The routine processing of business transactions is faster and cheaper than manual method.
4. It fastens the planning and control functions of management.
5. The information contents of MIS can be used by many managers for their varied needs.
6. It can be used to achieve competitive advantage in the marketplace.
7. Expected cost savings/reductions
8. Increase in output volumes and Increase in profit
9. Increase in accuracy level
10. Improvement in working environment
11. Improvement in turnaround time
12. Tangible benefits, e.g customer relations and achievement of set goals.

Characteristic features of MIS

- a) MIS supports structured decision at the operational and management control levels. However, they are also useful for planning purposes of senior management staff.
- b) MIS are generally reporting and control oriented. They are designed to report on existing operations and therefore to help to provide day-to-day control of operations.
- c) MIS rely on existing corporate data and data flows. d) MIS have little analytical capability.
- e) MIS generally used in decision making using past and present data. f) MIS is relatively inflexible.
- g) MIS has an internal rather than an external orientation.